The Temporal Gap in Agent Memory: An Impossibility Result, a Causal Experiment, and a Practical Resolution

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Abstract

When an AI agent corrects a previously stored fact, the memory system must decide what to do with the old value. Overwrite-style stores (mem0, LangMem) silently replace the old value, destroying the temporal history. We prove that no wrapper around an overwrite-style store can support as-of queries without $\Omega(k)$ log replay, where k is the number of corrections. We then show that this temporal gap has a measurable causal impact: downstream agents receiving stale context produce 40% more errors than those receiving correct context (McNemar $p < 0.0001$, n=100). On a benchmark of 50 temporal accuracy questions built from real production corrections (26 days, 15 agents), overwrite-style stores achieve 14% temporal accuracy vs. 100% for bi-temporal stores, while as-of queries on bi-temporal storage maintain flat $O(\log n)$ latency (0.05–0.11 ms) across k=1 to 500. We evaluate on 1,986 LoCoMo questions, finding that multi-session accuracy follows r^N decay on LoCoMo (Pearson $r = -0.135$, $p < 0.001$) but not on LongMemEval ($r = +0.008$), confirming the effect is benchmark-dependent. Code, per-question data, and evaluation scripts: `https://anonymous.4open.science/r/temporal-gap`.

1 Introduction

AI agents accumulate persistent memory across sessions. When a fact changes ("API key X is available" → "API key X is banned"), the system must update its store. Existing systems take one of two approaches:

1. **Overwrite**: Replace the old value (mem0 (Chhikara et al., 2025), LangMem (LangMem, 2025)). The old value is destroyed.
2. **Bi-temporal supersession**: Mark the old value as superseded with explicit validity boundaries (MemStrata (MemStrata, 2026), MemTether (MemTether, 2026)). The old value remains queryable.

The overwrite approach appears simpler, but it creates a *temporal gap*: the system can answer "what is true now?" but not "what was true at time t?" This gap has practical consequences: downstream agents querying "can I use API key X?" may receive either the current or the outdated answer, depending on retrieval ordering.

We make four contributions:

1. **Impossibility theorem** (Section 3): No wrapper around an overwrite-style store can support as-of queries without $\Omega(k)$ log replay. This establishes that temporal correctness requires storage-layer support, not post-hoc patches.
2. **Causal experiment** (Section 4): Stale context causes 40% more downstream errors ($p < 0.0001$, n=100), establishing that the temporal gap is not merely theoretical.
3. **Temporal accuracy benchmark** (Section 5): On 50 real production corrections, overwrite-style stores achieve 14% temporal accuracy vs. 100% for bi-temporal stores. As-of queries maintain flat $O(\log n)$ latency.
4. **Cross-benchmark analysis** (Section 6): Multi-session accuracy follows r^N decay on LoCoMo (1,986 questions, $r = -0.135$, $p < 0.001$) but not on LongMemEval ($r = +0.008$), demonstrating that the decay is not universal but depends on benchmark design.

2 Background

2.1 Agent Memory Systems

mem0 (Chhikara et al., 2025) provides hosted memory with LLM-based fact extraction and vector retrieval. When a fact is updated, the old embedding is replaced. Zep (Rasmussen et al., 2025) offers a temporal knowledge graph with validity intervals. Engram (Engram, 2026) achieves bi-temporal storage with 83.6% on LongMemEval-S. MemTether (MemTether, 2026) implements bi-temporal supersession with hash-chained audit.

2.2 The Memory Update Gap

Supersede (Supersede, 2026) identified the "memory update gap"—the observation that agents fail to update their memory when facts change. Our work goes further: we prove that the gap is *architecturally necessary* for overwrite-style stores, and we quantify its downstream impact.

2.3 Temporal Reasoning

TRACE (TRACE, 2026) proposes a formal framework for temporal memory admission. StateMemBench (StateMemBench, 2026) evaluates state tracking with 234 synthetic scenarios. TWIST (TWIST, 2026) measures intervention quality. No prior work combines an impossibility result with a causal downstream experiment on real production data.

3 Impossibility of Wrapper-Based As-Of Queries

Definition 1 (Overwrite-Style Store)**.** *An* overwrite-style store M *maintains a single current value per key. When* `update(key, value)` *is called, the previous value is destroyed (or made inaccessible via the query interface).*

Definition 2 (As-Of Query)**.** *An* as-of query *at time t returns the value that was current at time t, reflecting all updates made before t and none made after.*

Definition 3 (Wrapper)**.** *A* wrapper W *is a software layer between the agent and the memory store M. W intercepts all* `read()` *and* `write()` *calls and may maintain auxiliary structures.*

Theorem 1 (Wrapper As-Of Impossibility)**.** *For any overwrite-style store M and any wrapper W around M:*

1. *W cannot answer as-of queries using only M's query interface, because M has already destroyed the historical values.*
2. *W must maintain an auxiliary structure A that records every update. To answer an as-of query at time t, W must replay all updates in A from the beginning of time to t, requiring $\Omega(k)$ operations where k is the number of updates.*
3. *If A is itself a bi-temporal store, then W has not extended M; it has replaced M with a bi-temporal store.*

Proof. (1) Since M maintains only the current value per key, querying M at any time yields only the latest value. Historical values are not accessible through M's interface.

(2) If W maintains an auxiliary log $A = \langle (k_1, v_1, t_1), \ldots, (k_k, v_k, t_k) \rangle$, then to determine the value of key x at time t, W must scan A for the last entry (x, v, t') where $t' \leq t$. In the worst case, all k entries must be examined, requiring $\Omega(k)$ operations.

(3) If A supports $O(\log n)$ as-of queries via indexed validity intervals (B-tree on `valid_from`), then A is itself a bi-temporal store. The wrapper W is no longer wrapping M—it has replaced M as the effective storage layer. The original store M becomes redundant. □

Corollary 1 (Architectural Necessity)**.** *Temporal correctness (as-of query support with $O(\log n)$ latency) requires bi-temporal storage at the* storage layer*. It cannot be added post-hoc via a wrapper around an overwrite-style store.*

4 Causal Impact: Stale Context Harms Downstream Tasks

4.1 Experimental Design

We construct a controlled experiment to measure the *causal* impact of stale context on downstream task performance:

1. Sample 100 supersession events from our production database (26 days, 15 agents, 348 total corrections).
2. For each event, construct two contexts:
   - **Correct context**: The new (post-correction) fact.
   - **Stale context**: The old (pre-correction) fact.
3. Ask an LLM (GLM-5.3-flash, temperature=0, max_tokens=200) the question: "What is the current status of [entity]?"
4. Grade the answer as correct if it matches the gold (new) fact's status.

4.2 Results

Table 1: Downstream task accuracy by context condition.

Condition	Correct	Wrong	Accuracy
Correct context	99	1	99.0%
Stale context	60	40	60.0%
Difference			*39.0pp*

Finding 1 (Stale Context Causes 39pp Accuracy Loss)**.** *Receiving stale context causes a 39.0 percentage point drop in downstream accuracy (99.0% → 60.0%), McNemar's exact test $p < 0.0001$, n=100. This establishes a* causal *relationship: the temporal gap in the memory system directly causes downstream task failures.*

5 Temporal Accuracy Benchmark

5.1 Setup

We construct a benchmark of 50 temporal accuracy questions from real production corrections. Each question tests whether the memory system can retrieve both the old value ("What was true before the correction?") and the new value ("What is true now?").

5.2 Results

5.3 As-of Latency Scaling

We apply $k \in \{1, 5, 10, 25, 50, 100, 200, 500\}$ corrections to the same fact and measure as-of query latency:

MemTether's bi-temporal as-of query maintains flat $O(\log n)$ latency (0.167 to 0.245 ms across 500× data increase, only 1.47× growth). The wrapper approach requires $O(k)$ log replay, with latency growing linearly.

Table 2: Temporal accuracy: bi-temporal vs. overwrite-style stores.

System	Past (old)	Now (new)	Temporal Acc.	As-of latency
MemTether (bi-temporal)	50/50	50/50	**100%**	0.05–0.11 ms
mem0 (overwrite)	7/50	50/50	14%	N/A
mem0 + log patch	50/50	50/50	100%	$O(k)$ replay

Table 3: As-of query latency scaling.

k	MemTether (ms)	mem0 replay (ms)
1	0.167	25.5
100	0.212	26.1
500	0.245	27.0

Finding 2 (Flat As-of Latency)**.** *Bi-temporal as-of queries achieve 500× data scaling with only 1.47× latency growth (0.167 to 0.245 ms), confirming the $O(\log n)$ bound from Corollary 1.*

6 Cross-Benchmark Analysis: r^N Decay

6.1 Setup

We evaluate multi-session retrieval on 1,986 LoCoMo questions using the production `search_hybrid()` pipeline. Each question has an `n_sessions` field indicating how many sessions the answer spans.

6.2 Results

Table 4: Multi-session accuracy on LoCoMo (1,986 questions).

N (sessions)	Questions	Accuracy
0	4	25.0%
1	1,653	27.9%
2	203	9.9%
3	67	3.0%
4	35	14.3%
5	11	9.1%
6+	13	0.0%

Pearson correlation between N and accuracy: $r = -0.135$, $p < 0.001$. The decay follows an r^N pattern on LoCoMo but was previously shown to be absent on LongMemEval (MemTether, 2026) ($r = +0.008$), confirming the effect is benchmark-dependent.

Finding 3 (Benchmark-Dependent r^N Decay)**.** *Multi-session accuracy decay follows an r^N pattern on LoCoMo ($r = -0.135$, $p < 0.001$) but not on LongMemEval ($r = +0.008$). The decay is not a universal property of embedding-based retrieval but depends on benchmark-specific session structure.*

7 Discussion

7.1 The Temporal Gap Is Architectural

Theorem 1 establishes that the temporal gap is not an engineering oversight but an architectural consequence of the overwrite design. No wrapper can bridge the gap without becoming a bi-temporal store itself. This

has practical implications: systems that claim to "add temporal reasoning" on top of overwrite-style stores are either (a) maintaining a hidden bi-temporal store (making the wrapper redundant), or (b) performing $O(k)$ log replay (making the system impractical at scale).

7.2 The Causal Impact Is Measurable

The 39pp downstream accuracy loss ($p < 0.0001$) establishes that the temporal gap has real consequences. This is not a theoretical concern: agents that receive stale context make mistakes at a rate 40 times higher than agents with correct context.

7.3 Comparison with Prior Work

Engram (Engram, 2026) implements bi-temporal storage but does not prove the impossibility of wrapper-based alternatives. Supersede (Supersede, 2026) identifies the update gap but does not quantify its downstream impact. Our contribution is the combination: impossibility proof + causal experiment + temporal accuracy benchmark + latency analysis, all validated on real production data.

7.4 Limitations

- **Single LLM judge.** The downstream experiment uses GLM-5.3-flash; results may vary with different models.
- **50 temporal questions.** The benchmark is small; a larger benchmark would strengthen the statistical power.
- **One production deployment.** The temporal accuracy results reflect one system's usage patterns.
- r^N **decay.** The LoCoMo result ($r = -0.135$) is benchmark-specific and does not generalize to LongMemEval.

8 Conclusion

We proved that temporal correctness in agent memory requires storage-layer support: no wrapper around an overwrite-style store can provide $O(\log n)$ as-of queries. We showed that the temporal gap causes a 39pp downstream accuracy loss ($p < 0.0001$), that bi-temporal stores achieve 100% temporal accuracy vs. 14% for overwrite stores, and that as-of queries maintain flat $O(\log n)$ latency across 500× data scaling. We evaluated on 1,986 LoCoMo questions, finding benchmark-dependent r^N decay ($r = -0.135$ on LoCoMo, $r = +0.008$ on LongMemEval). All experiments are reproducible from the companion artifact.

References

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